

Config Drift Analysis Report

Automated DevOps Compliance & Risk Assessment

Scan Timestamp	2026-06-09 12:49:04
Total Config Files Evaluated	1 (actual_config.json)
Total Anomalies Detected	8 issues
Risk Severity Profile	Critical: 2 High: 1 Medium: 5 Low: 0

Executive Summary

This configuration compliance audit reports drifts identified between intended configuration layouts and actual parameters retrieved from live production nodes. Discrepancies are categorized according to logical keyword threat scores (e.g., firewall parameters raise Critical ratings). Each active drift has been run through an AI impact analysis engine powered by Groq LLM. Engineering and Operations staff should review security-impacting modifications immediately to mitigate environment breaches, service outages, or compliance regressions.

Detailed Drift Registry

The table below lists every individual structural or parameter difference found during execution.

File Name	Config Path	Expected	Actual	Severity
actual_config.json	cache.enabled	true	false	Medium
actual_config.json	cache.ttl_seconds	600	Missing	Medium
actual_config.json	logging_level	INFO	DEBUG	Medium
actual_config.json	replicas	3	2	Medium
actual_config.json	version	1.2.0	1.2.1	Medium
actual_config.json	database.pool_size	20	10	High
actual_config.json	security.auth_method	oauth2	basic	Critical
actual_config.json	security.enable_tls	true	false	Critical

AI-Powered Impact & Remediation Review

Detailed technical audits and fix playbooks generated by Groq LLM for each configuration drift.

Issue #1: actual_config.json → cache.enabled

Drift Type: Modified Value | Severity: Medium | Expected: true | Deployed: false

Technical Impact: Disabling cache may lead to increased load times, reduced application performance, and potential capacity issues due to repeated database queries. This could also impact deployment health by introducing inconsistencies in application behavior.

Business Impact: End users may experience slower application performance, leading to decreased user satisfaction and potential revenue loss. Additionally, this configuration drift may impact SLAs and brand compliance by failing to meet performance expectations.

Security Impact: Disabling cache does not directly impact data protection or introduce vulnerabilities. However, it may increase the attack surface by making it easier for attackers to exploit application performance issues.

Remediation Plan:

To correct the drift, update the cache.enabled configuration to true, and verify that the application is functioning as expected. Additionally, monitor application performance and capacity to ensure that the cache is properly configured and functioning as intended.

Issue #2: actual_config.json → cache.ttl_seconds

Drift Type: Missing Key | Severity: Medium | Expected: 600 | Deployed: Missing

Technical Impact: The missing cache.ttl_seconds configuration may lead to cache data being stored indefinitely, causing increased memory usage and potential system crashes. This can impact system capacity and deployment health, resulting in decreased performance and increased downtime.

Business Impact: The prolonged cache data storage may lead to decreased application performance, affecting end-user experience and potentially resulting in lost revenue. Additionally, the increased memory usage may lead to higher costs for infrastructure upgrades and maintenance.

Security Impact: The missing cache.ttl_seconds configuration may expose sensitive data to unauthorized access, as cache data may not be properly purged. This increases the risk of data breaches and cyber attacks.

Remediation Plan:

Update the cache.ttl_seconds configuration to 600 seconds to ensure proper cache data purging. This can be achieved by running the following command: "config set cache.ttl_seconds 600". Additionally, consider implementing a cache data retention policy to ensure data is properly purged on a regular basis.

Issue #3: actual_config.json → logging_level

Drift Type: Modified Value | **Severity:** Medium | **Expected:** INFO | **Deployed:** DEBUG

Technical Impact: The logging level drift from INFO to DEBUG may lead to increased log volume, potentially impacting system performance, storage capacity, and deployment health. This could result in slower system response times, increased resource utilization, and potential errors due to log flooding.

Business Impact: The increased log volume may lead to higher storage costs, increased administrative overhead for log management, and potential SLA breaches due to system performance degradation. Additionally, the drift may compromise brand compliance by exposing sensitive information in debug logs.

Security Impact: The DEBUG logging level may expose sensitive information, such as API keys, passwords, or other confidential data, increasing the risk of data breaches and cyber attacks. This drift may also reveal internal system configurations, potentially aiding attackers in exploiting vulnerabilities.

Remediation Plan:

Correct the logging level drift by updating the logging configuration to the expected value of INFO. Implement a log rotation policy to manage log volume, and consider implementing log encryption and access controls to protect sensitive information.

Issue #4: actual_config.json → replicas

Drift Type: Modified Value | **Severity:** Medium | **Expected:** 3 | **Deployed:** 2

Technical Impact: The under-provisioned replicas may lead to increased latency, reduced system responsiveness, and potential service outages due to insufficient capacity. This could impact deployment health and necessitate manual intervention to scale up resources.

Business Impact: End users may experience decreased system performance, increased error rates, and reduced overall system reliability. This could result in increased support requests, decreased customer satisfaction, and potential revenue loss due to SLA breaches.

Security Impact: The under-provisioned replicas may increase the attack surface, making it more challenging to detect and respond to security incidents. Additionally, the reduced capacity may lead to data loss or corruption in the event of a security breach.

Remediation Plan:

To correct the drift, increase the number of replicas to the expected value of 3. This can be achieved by scaling up the resource pool or adjusting the deployment configuration to ensure sufficient capacity for the expected workload.

Issue #5: actual_config.json → version

Drift Type: Modified Value | Severity: Medium | Expected: 1.2.0 | Deployed: 1.2.1

Technical Impact: The configuration drift may cause unexpected behavior or errors in systems, potentially leading to capacity issues and deployment health problems. This could result in service downtime, delayed updates, or inconsistent performance.

Business Impact: The end users may experience service disruptions, delayed updates, or inconsistent performance, affecting their productivity and overall experience. This could also lead to increased costs due to manual intervention, extended support, or potential revenue loss.

Security Impact: The configuration drift may introduce security vulnerabilities or data protection risks if the updated version (1.2.1) contains security patches or fixes not present in the expected version (1.2.0). This could increase the risk of cyber attacks or data breaches.

Remediation Plan:

Update the configuration to the expected version (1.2.0) to prevent potential technical, business, and security issues. Review the change history and documentation to ensure no critical security patches or fixes are missed.

Issue #6: actual_config.json → database.pool_size

Drift Type: Modified Value | Severity: High | Expected: 20 | Deployed: 10

Technical Impact: The reduced database pool size from 20 to 10 may lead to increased latency, reduced system responsiveness, and potential deployment failures due to insufficient resources. This could impact application performance, user experience, and overall system capacity.

Business Impact: This configuration drift may result in decreased end-user satisfaction, potential revenue loss due to system downtime, and non-compliance with Service Level Agreements (SLAs). Additionally, it may damage the company's brand reputation if customers experience frequent system failures.

Security Impact: The reduced database pool size does not directly impact data protection or introduce new vulnerabilities. However, it may increase the risk of cyber attacks if the system is unable to handle increased traffic or requests, leading to potential data breaches or unauthorized access.

Remediation Plan:

Increase the database pool size to 20 to ensure optimal system performance, user experience, and compliance with SLAs. Monitor system resources and adjust the pool size as needed to prevent future configuration drift.

Issue #7: actual_config.json → security.auth_method

Drift Type: Modified Value | Severity: Critical | Expected: oauth2 | Deployed: basic

Technical Impact: The configuration drift from 'oauth2' to 'basic' auth method may lead to unauthorized access to sensitive systems and data. This could result in compromised system integrity, data breaches, and potential downtime due to security incidents.

Business Impact: The use of 'basic' auth method may lead to non-compliance with security standards, resulting in reputational damage, financial losses due to data breaches, and potential legal liabilities. End-users may experience decreased system availability and reliability.

Security Impact: The 'basic' auth method is a weak authentication mechanism that exposes the environment to brute-force attacks, phishing, and other social engineering tactics. This may lead to unauthorized access to sensitive data, including user credentials, financial information, and confidential business data.

Remediation Plan:

To correct the configuration drift, update the 'security.auth_method' to 'oauth2' and ensure that all systems and services are reconfigured to use the new authentication method. Additionally, conduct a thorough security audit to identify and remediate any potential vulnerabilities.

Issue #8: actual_config.json → security.enable_tls

Drift Type: Modified Value | Severity: Critical | Expected: true | Deployed: false

Technical Impact: Disabling TLS (Transport Layer Security) may expose sensitive data in transit to eavesdropping, tampering, and man-in-the-middle attacks. This can lead to system instability, capacity issues, and deployment health problems due to potential security breaches.

Business Impact: End users may experience data breaches, compromised sensitive information, and potential financial losses. Cost implications may arise from security incidents, and SLAs may be compromised due to system downtime. Brand reputation may also be affected by security vulnerabilities.

Security Impact: Data protection is severely compromised, and the system becomes vulnerable to various cyber attacks, including eavesdropping, tampering, and man-in-the-middle attacks. Disabling TLS also increases the risk of data breaches, unauthorized access, and potential security incidents.

Remediation Plan:

To correct the drift, update the security configuration to enable TLS (security.enable_tls = true). This can be achieved by running the following commands: 1. Update the security configuration file to set security.enable_tls to true. 2. Restart the affected services to apply the changes. 3. Verify that TLS is enabled and functioning correctly.